How Does a Coding Agent Work?

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MVP Microsoft Most Valuable Professional

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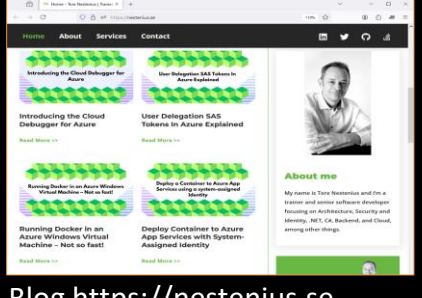
About Tore Nestenius

Work https://tn-data.se



Blog https://nestenius.se

.NET Skåne

https://meetup.com/net-skane

Have a question about this session or beyond?

tore@tn-data.se

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First, One Important Thing

Do Ask Questions!

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The AI Revolution

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The AI Revolution

Claude skapar kaos – konsultfirma slutar ta betalt för kodning: “Oundvikligt”

WIZARDWORKS SAAS AI 28 jan 2026, kl 11:26

https://www.breakit.se/artikel/45281/claude-skapar-kaos-konsultfirma-slutar-ta-betalt-for-kodning-oundvikligt

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5

The AI Revolution

Rebel and Bird
1,882 followers
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3d •

Är konsulttimmen död?
Vi tror det. Den traditionella modellen där man säljer tid istället för värde har nått vägs ände.
Därför genomför vi på Rebel and Bird nu vår största transformation någonsin. Vi ställer om hela verksamheten till Agentic First.
Det innebär att vi lämnar löpande räkning bakom oss och lanserar Output-As-A-Service. Genom att kombinera våra seniora konsulter med autonoma AI-agenter kan vi nu erbjuda en ekvation som tidigare var omöjlig:
-Svensk spetskompetens och fysisk närvaro.
-Korta ledtider och tät dialog.
-En prislapp som matchar offshore-leverans. Till en fast månadskostnad.

Varför välja mellan kvalitet och pris när tekniken nu tillåter båda?
Vi bygger framtiden istället för att vänta på den. Är ni redo att sluta köpa timmar och börja prenumerera på resultat?

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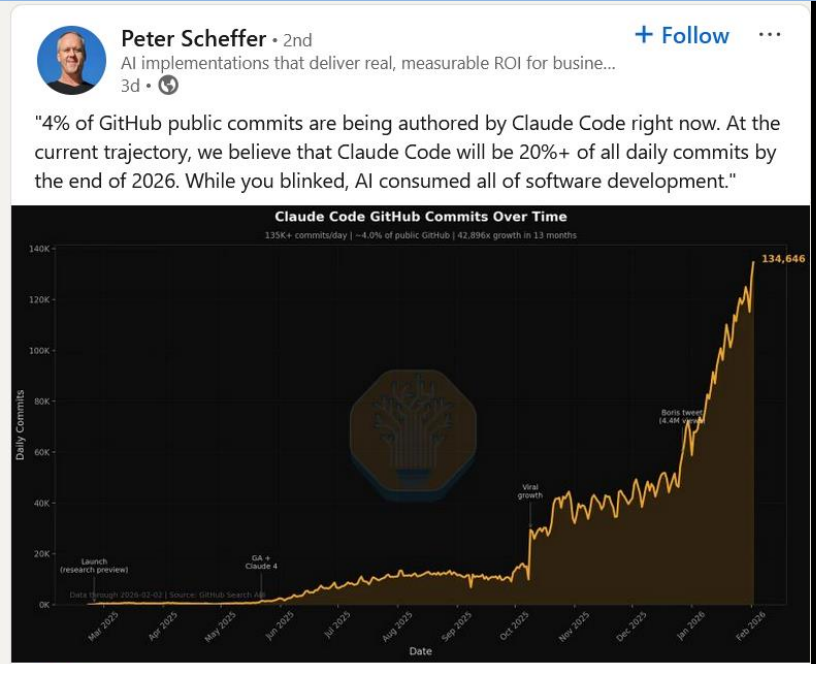
The AI Revolution



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How many of you…

Claude Code, Copilot, Cursor…

Have used a **Coding Agent**?

Could **explain** how it actually works?

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Why Did I Write This Talk?

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Why Did I Write This Talk?

I want to teach Agentic AI Development

But how should I teach it?

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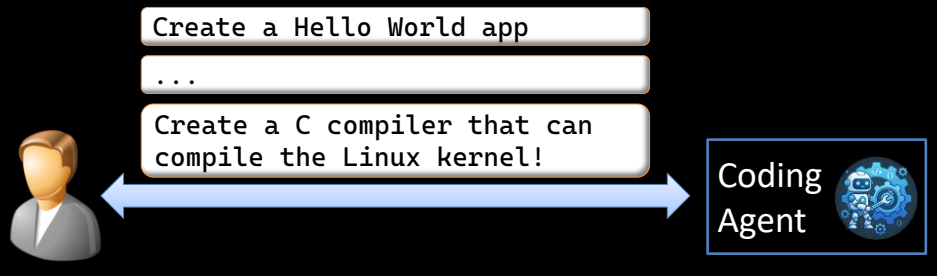
Why Did I Write This Talk?

Just writing prompts is not enough!



This will not teach you agentic coding!

What do we need instead?

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What do we need instead?

We need Mechanical Sympathy!

- Understand what happens **under the hood**
- Know how the agent **talks** to the LLM
- Understand **tools** and how the agent uses them
- Learn what drives token **usage** and **cost**

You don't drive a race car by ignoring the engine

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12

What Is Agentic Coding?

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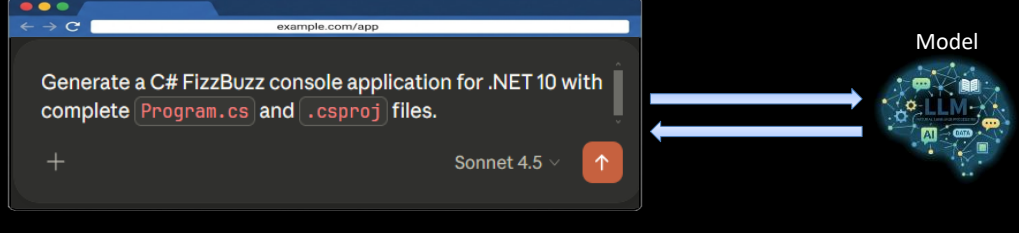
13

Introduction

Most of us started by using LLMs in the browser



Why doesn't this scale to real software work?

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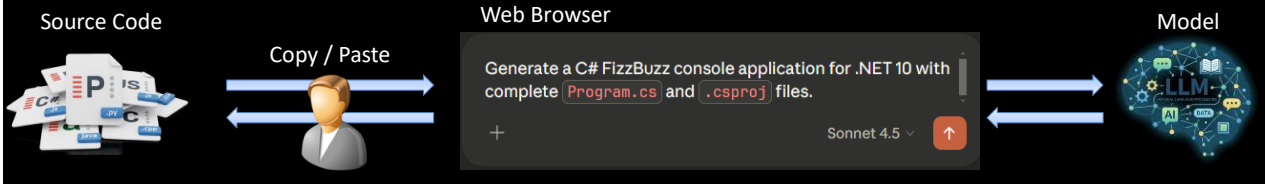
Introduction

The developer becomes the message transport



Copy -> Paste -> Error -> Copy -> Paste -> Repeat...

Can we remove the human from the feedback loop?

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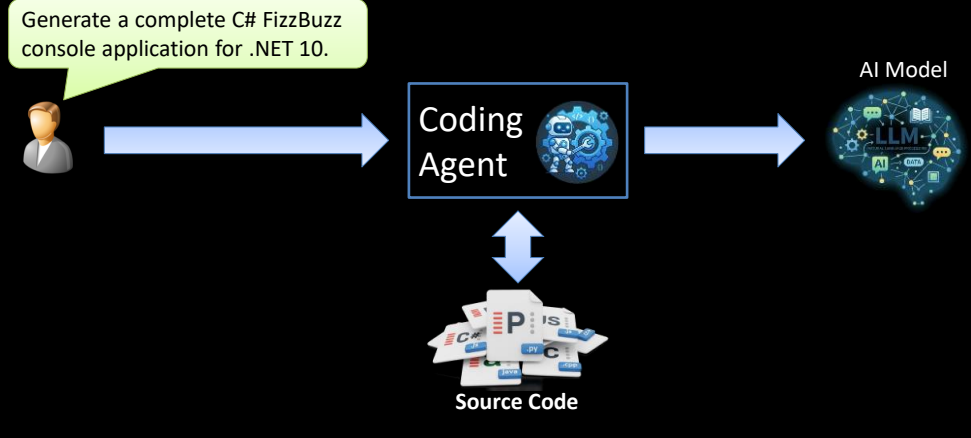
What Is Agentic Coding?

Agentic Coding: the AI operates directly on your codebase



So, how does this work?

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16

Coding Agents

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Coding Agents

Meet your new coding partner: The Coding Agent

How can it access my source code files?

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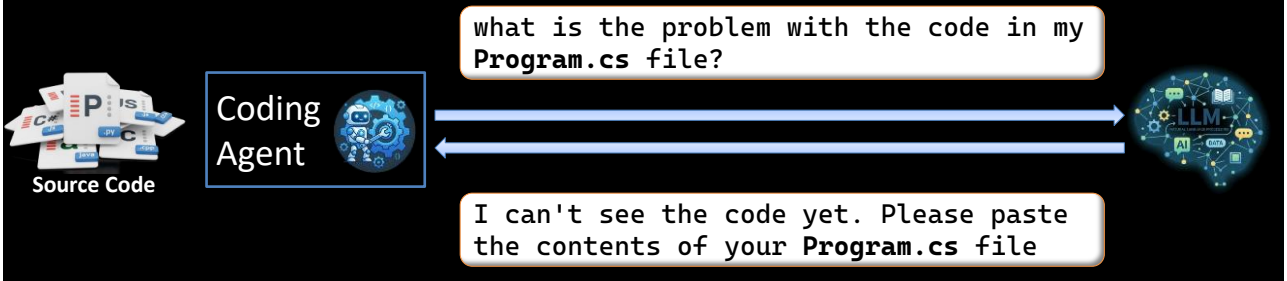
Coding Agents

Large Language Models (LLMs) operate only on text



How can we observe this interaction?

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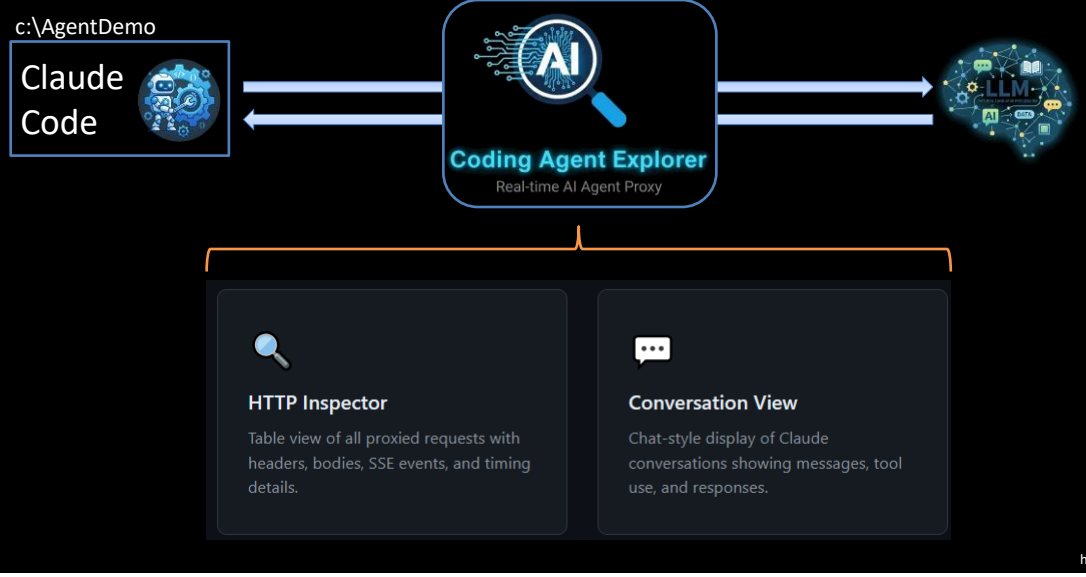
Demo #1



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How do we give it access to the source code?

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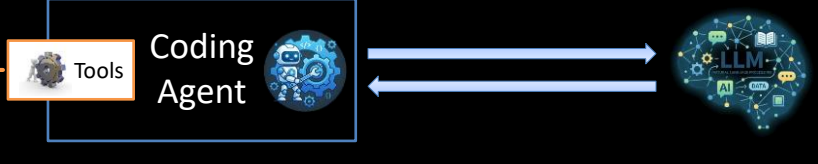
Tools

We can add a set of **tools** to our coding agent

- File I/O
- Run commands
- Web fetch
- Web search
- …



How can the LLM use these tools?

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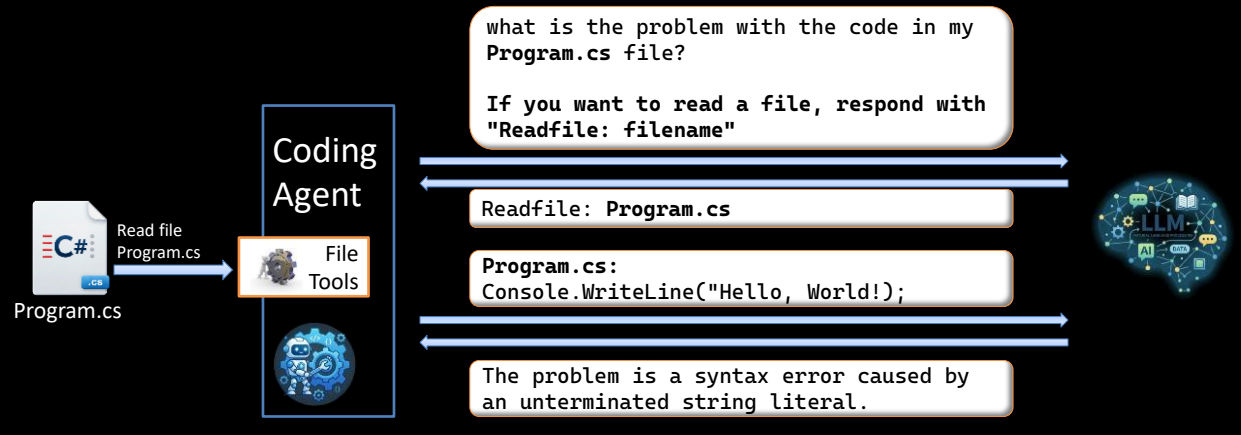
Tools

The coding agent describes the available tools to the LLM



All coding agents work in a similar way!

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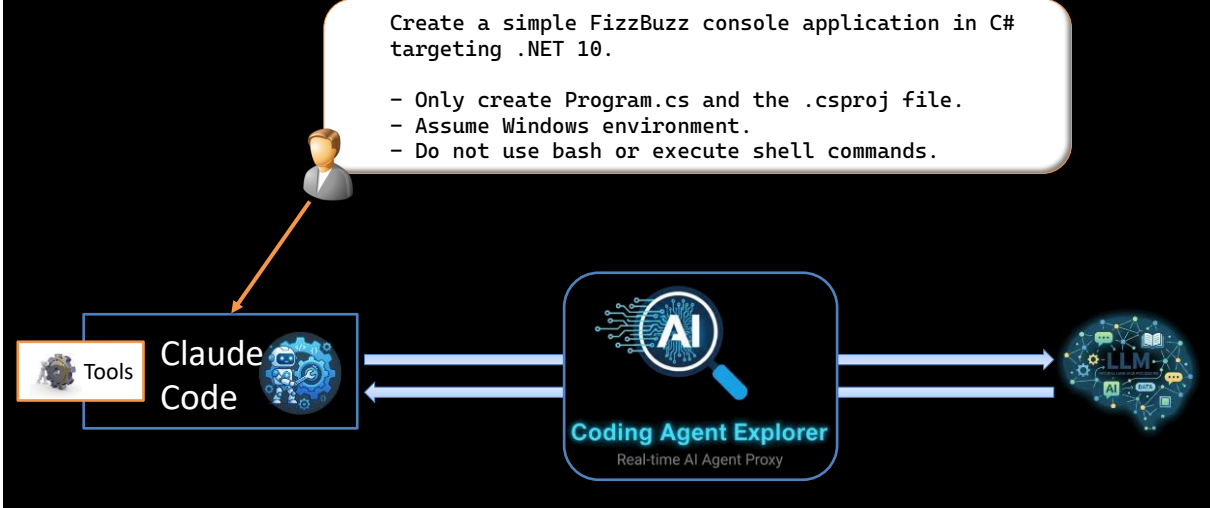
Demo #2 – Exploring the tools



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Model Context Protocol (MCP)

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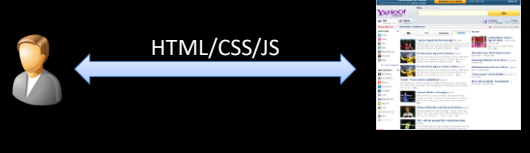
MCP

First, we had the web, for humans to consume



Then came the need for APIs

If we already have APIs, why do we need MCP?

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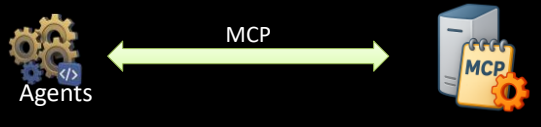
MCP

APIs are designed for applications, not agents

MCP lets you expose services and tools to agents



What does MCP specify?

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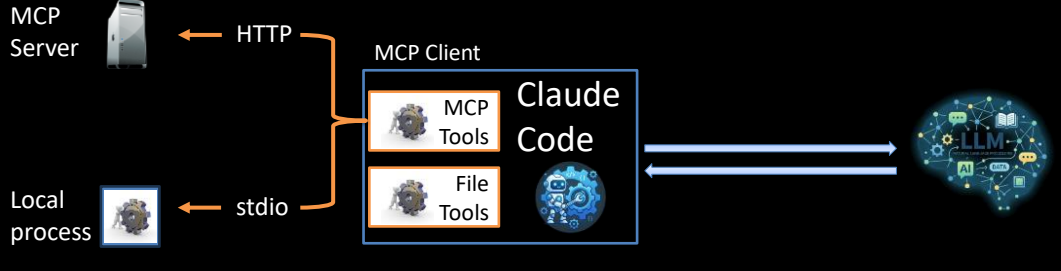
MCP Tools

MCP specifies the protocol between an agent and server



How do we use these MCP tools in our coding agent?

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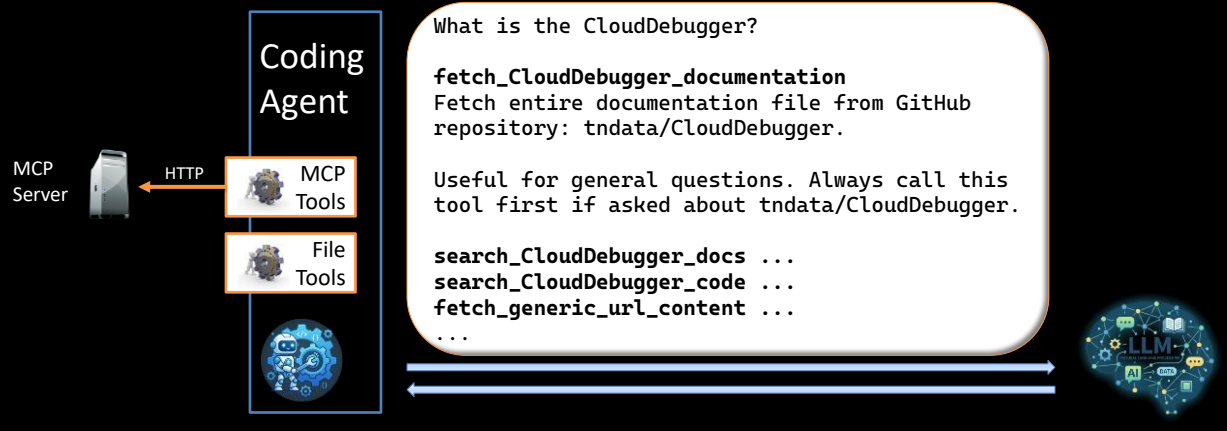
MCP Tools

MCP tools are exposed to the model as regular tools



The model never communicates directly with MCP servers

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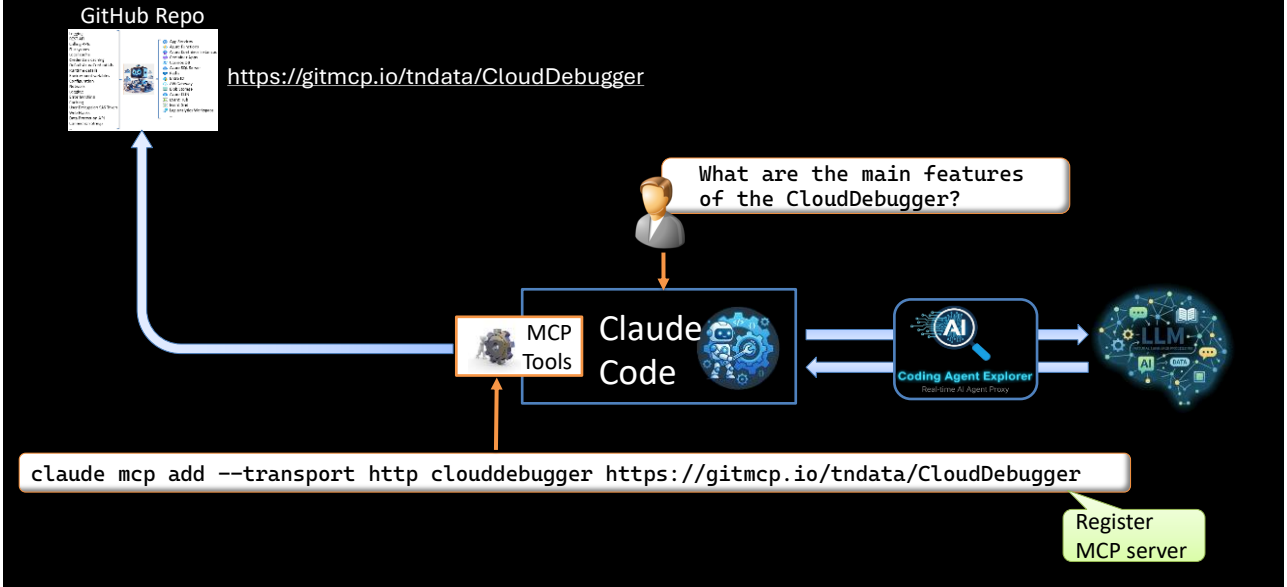
Demo #3 – Exploring MCP



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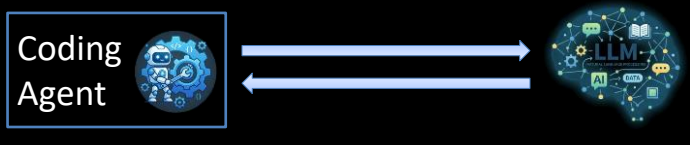
The Three Pieces of an LLM Request



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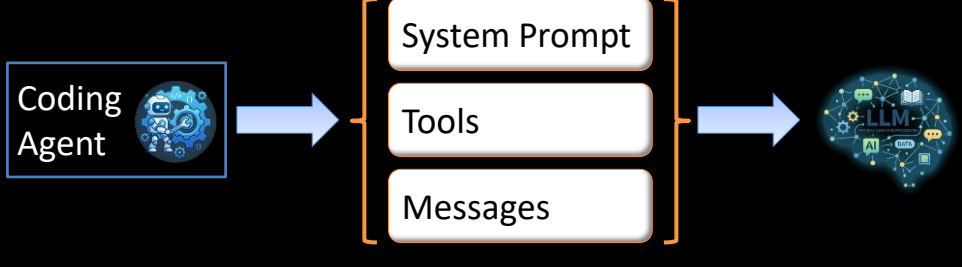
The Three Pieces of an LLM Request

Every LLM request consists of three parts:



What are the differences?

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The Three Pieces of an LLM Request

System Prompt: The LLM's instruction manual

Coding Agent

```
You are an expert C# coding assistant.

Your role is to help developers write, debug, and
improve C# code.

Guidelines:
- Use modern C# features and best practices
- Follow SOLID principles
- Write clean, maintainable code
- Explain your reasoning when making changes
- Always verify your solutions work correctly


Be concise, helpful, and focus on producing working
code.
```

Included in every request

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33

The Three Pieces of an LLM Request

Tools: The list of available tools that the LLM can call

Coding Agent

```
Task
TaskOutput
Bash
Glob
Grep
ExitPlanMode
Read
Edit
Write
NotebookEdit
WebFetch
WebSearch
TaskStop
AskUserQuestion
Skill
EnterPlanMode
TaskCreate
TaskGet
TaskUpdate
TaskList
ToolSearch
```

```
Executes a given bash command with optional
timeout. Working directory persists between
commands; shell state (everything else) does not.
The shell environment is initialized from the
user's profile (bash or zsh).

IMPORTANT: This tool is for terminal operations
like git, npm, docker, etc. DO NOT use it for file
operations (reading, writing, editing, searching,
finding files) - use the specialized tools for
this instead.
...
```

Included in every request

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34

The Three Pieces of an LLM Request

Messages: The ongoing conversation

Coding Agent

```
what is the problem with the code in my
Program.cs file?

If you want to read a file, respond with
"Readfile: filename"
```

Included in every request

```
Readfile: Program.cs
```

```
Program.cs:
Console.WriteLine("Hello, World!);
```

```
The problem is a syntax error caused by
an unterminated string literal.
```

```
Thanks! You are a lifesaver!
```

The latest prompt

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35

Demo #4 – Exploring the request

```
Plan Mode

Sonnet 4.5 | Context: 10% used | Cost: $0,05 | AgentDemo
plan mode on (shift+tab to cycle)
```

What does **Plan Mode** do internally?

Claude Code

Coding Agent Explorer

Real-time AI Agent Proxy

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36

Hooks

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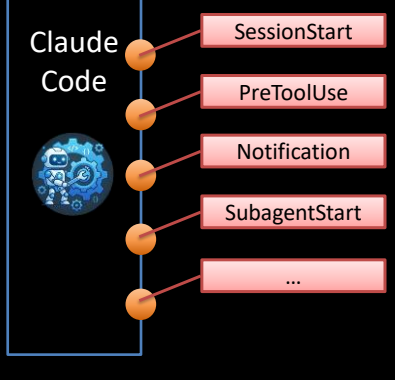
37

Hooks

Hooks run commands when Claude Code events occur



What can hooks be used for?

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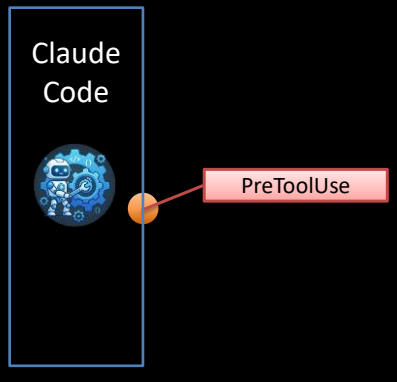
Hooks

What you can do with hooks



With hooks, you can:

- Enforce security policies before tools run
- Log and audit tool usage
- Trigger external automation
- Notify external systems (Slack, CI, etc.)

Which events can we hook into?

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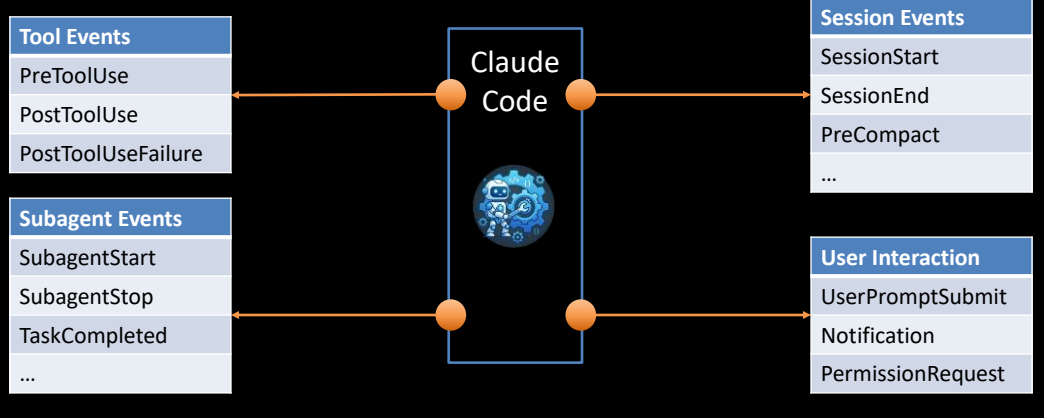
Hooks

Claude Code exposes four categories of **hookable events**



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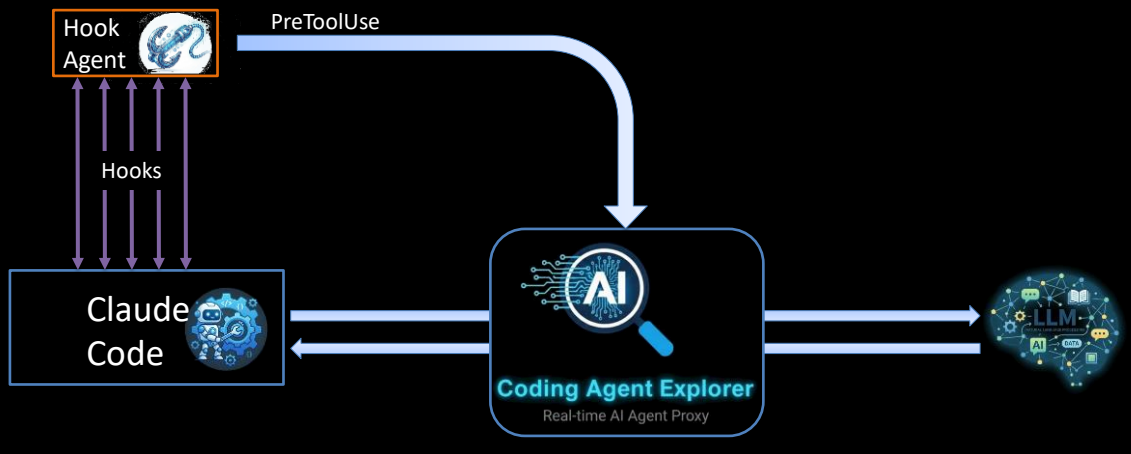
Demo #5 – Exploring Hooks



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41

Building Your Own Agent in C#

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42

Building Your Own Agent

An agent is just a loop:

```
var agent = new Agent();

while (true)
{
    var input = GetUserInput();

    var response = agent.SendMessage(input);

    Print(response);
}
```

What about the message history (context)?

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Building Your Own Agent

The context is just a collection

```
var context = new ContextCollection();

while (true)
{
    var input = GetUserInput();

    context.AddUserPrompt(input);

    var response = agent.SendMessage(context);

    context.AddAgentResponse(response);

    Print(response);
}
```

What about tool support?

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44

Building Your Own Agent

Adding tool support is easy:

```
while (true)
{
    var input = GetUserInput();

    context.AddUserPrompt(input);

    var response = agent.SendMessage(context);

    // Tool chain: read → edit → test
    while (agent.WantsToUseTool(response))
    {
        // Executes tool + appends tool result to context
        response = agent.ProcessToolCall(context, response);
        response = agent.SendMessage(context);
    }

    context.AddAgentResponse(response);
    Print(response);
}
```

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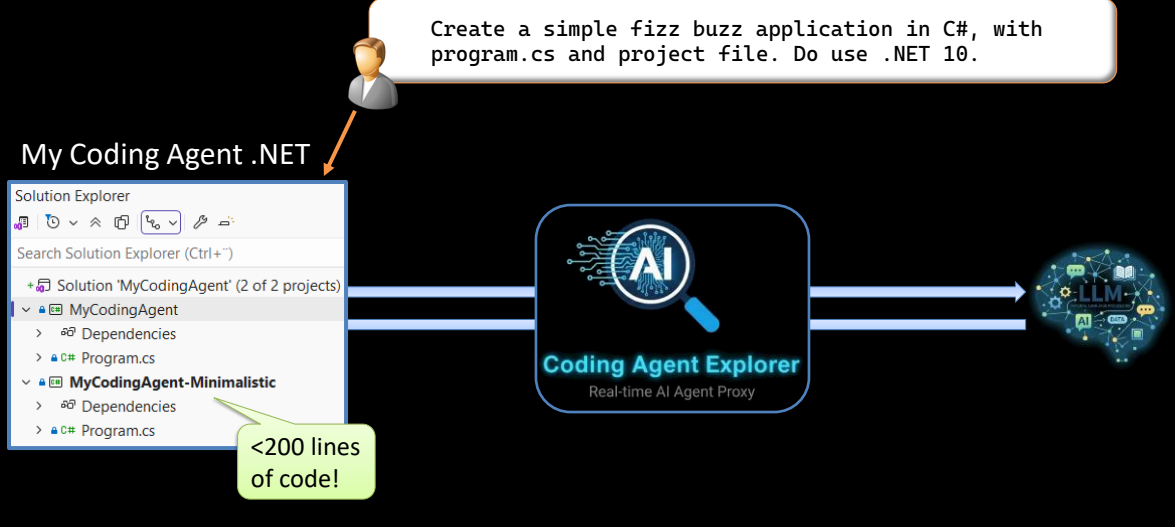
Demo #6 – MyCodingAgent.NET



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Questions?

Your next steps:

Scan for slides + Resources

- ✅ Book this presentation for my company
- ✅ Run the Coding Agent Explorer
- ✅ Follow Tore on LinkedIn

github.com/tndataab/PublicBlogContent

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